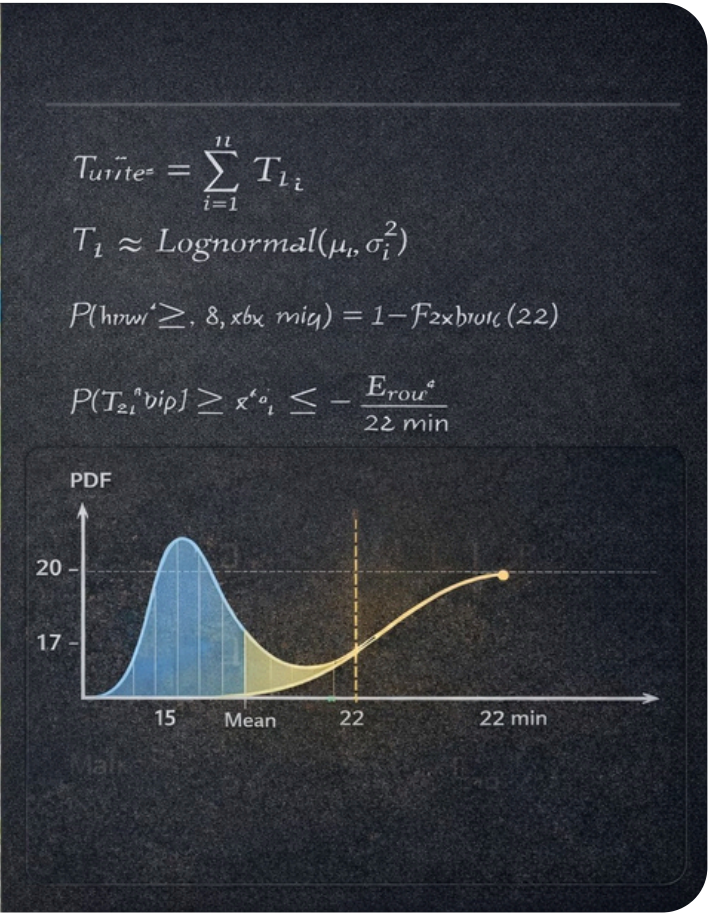
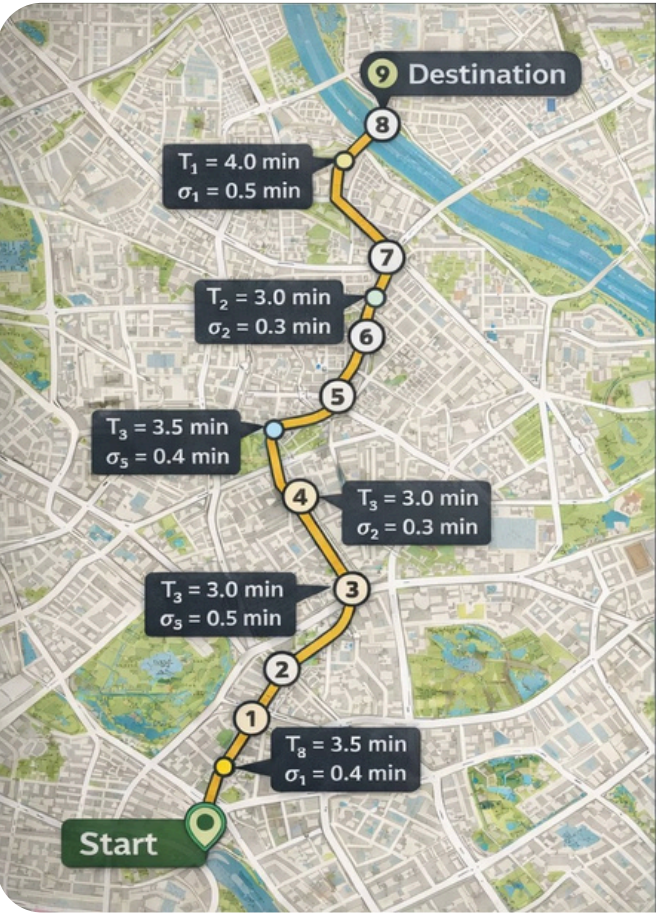


Probabilistic Route Selection for On-Time Arrival

Fundamentals of Probability in Computing | CSE 400



Project Context and Motivation



The Challenge

Intelligent transportation systems require reliable route planning to optimize travel efficiency. However, traffic conditions are inherently uncertain4travel times fluctuate due to demand variations, incidents, and congestion.

The Gap

Traditionaldeterministic shortest-path routing assumes fixed travel times, which may fail when real-world conditions are stochastic.

Our Focus

Analyzing how deterministic routing performs under realistic uncertainty.

Project Objective and Baseline Strategy

Primary Objective

Evaluate reliability of deterministic routes under stochastic travel time conditions

Baseline Strategy

Fixed route selection using A* shortest path algorithm minimizing expected travel time

Decision Rule

Route never changes during simulation
only travel times vary between runs

Why deterministic routing? It's computationally efficient, widely deployed in production systems, and provides a meaningful performance baseline for comparison with more sophisticated strategies.

System Pipeline Overview



This simulation framework models realistic traffic uncertainty while maintaining a fixed routing decision, enabling systematic analysis of deterministic strategy performance.

Sources of Uncertainty

Traffic Demand Fluctuations

Peak hours, special events, and seasonal patterns cause demand to vary significantly

Vehicle Interactions

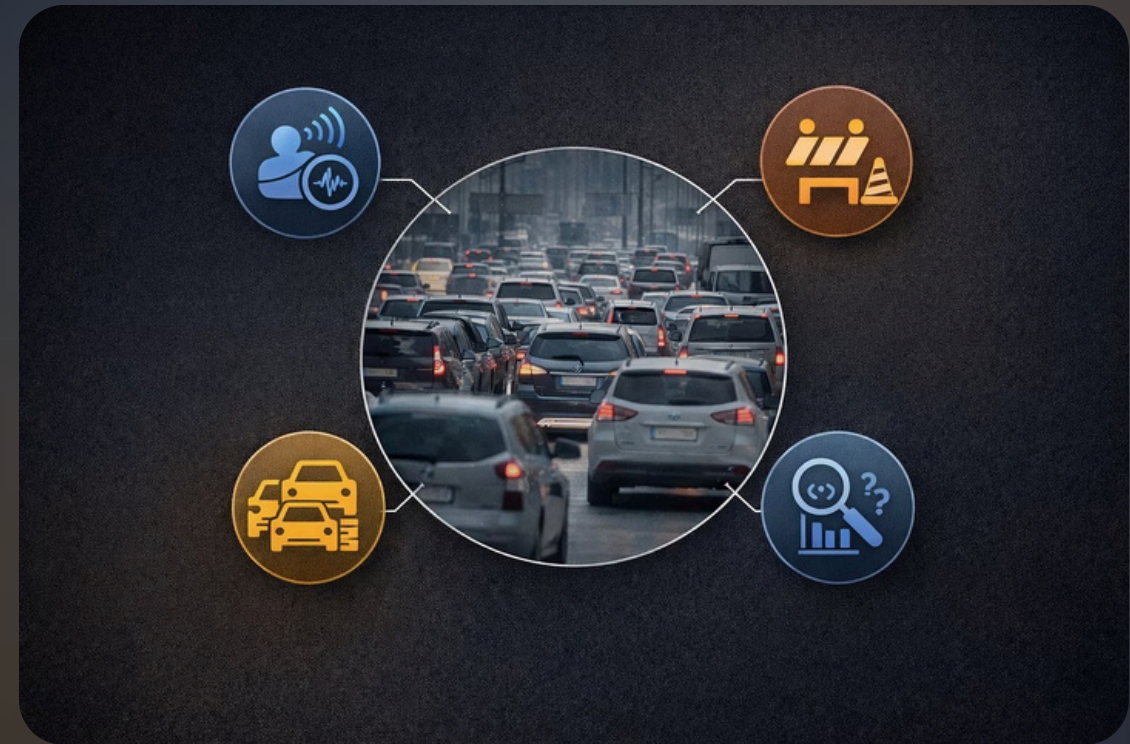
Merging, lane changes, and driver behavior create unpredictable delays

Incidents and Congestion

Accidents, construction, and bottlenecks introduce sudden travel time increases

Modeling Assumptions

Aggregate representations and simplified physics ignore micro-level complexity



Key Random Variables

Segment Travel Time

T_i represents travel time for road segment i

Total Route Time

$T_{route} = \sum_i T_i$

Arrival Deviation

$A = T_{route} - D$

$A < 0$: Early arrival

$A > 0$: Late arrival

$A = 0$: On-time arrival

Probabilistic Modeling

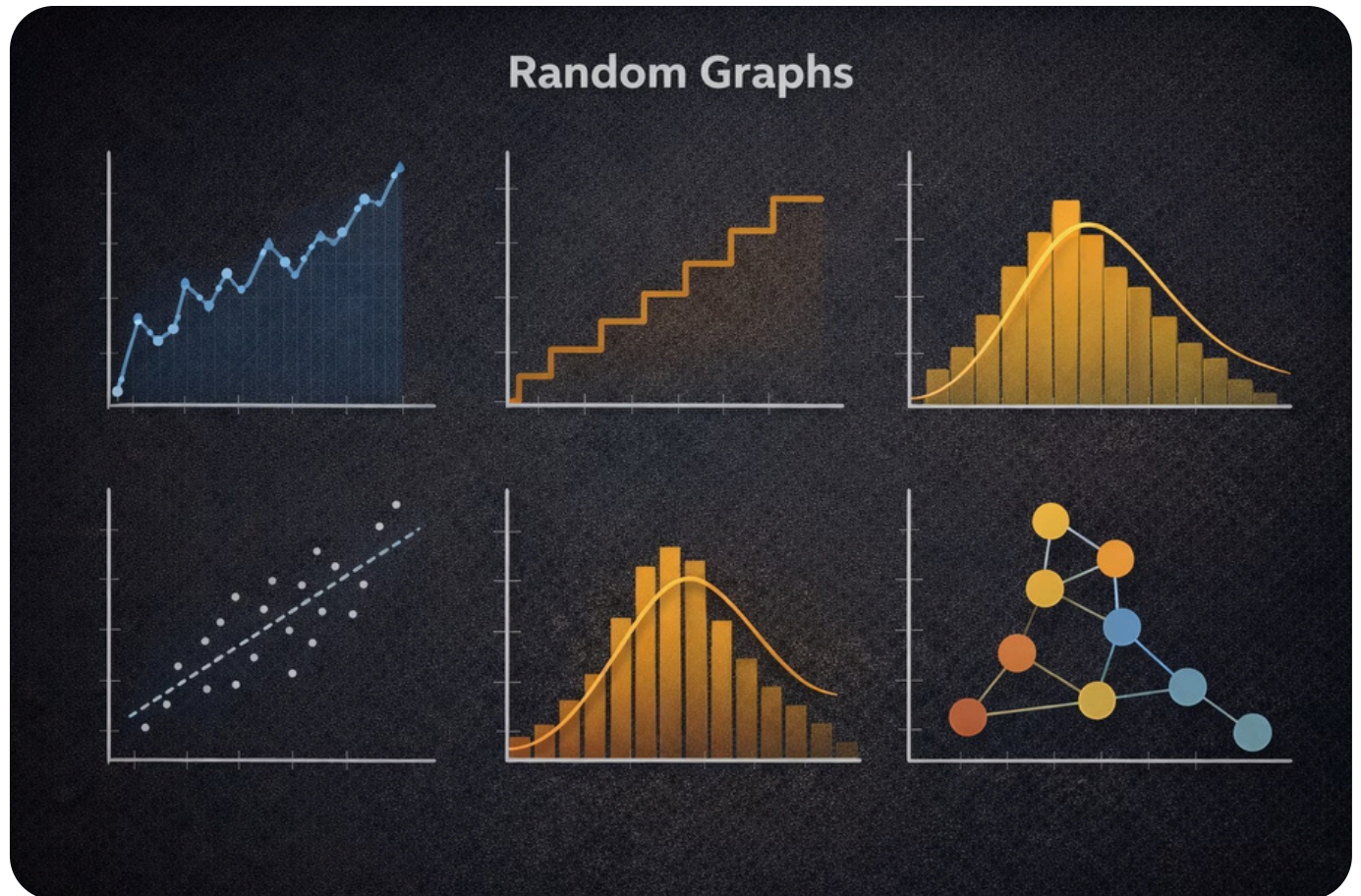
Model Choice

Travel time for segment i follows lognormal distribution:

$$T_i \sim \text{Lognormal}(i, \tilde{A}_i^2)$$

Why Lognormal?

- Travel time always positive
- Right-skewed distributions match traffic delay patterns
- Sum of lognormal approximates route time distribution



Stochastic inputs propagate through the route summation, creating variable total travel times across simulation runs.

Deterministic Decision Rule



Pre-computation

A*algorithm computes route using expected edge weights based on mean travel times



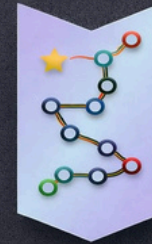
Fixed Selection

Chosen route remains unchanged during all simulation runs



Environmental Randomness

Only travel times vary while decision rule is deterministic



A* Algorithm

Simulation Implementation



Generate Travel Times

Sample random travel times for each road segment from lognormal distributions



Select Route

A* algorithm computes deterministic route using expected edge weights



Sample & Simulate

For each simulation run k , sample travel times for route segments



Compute Total Time

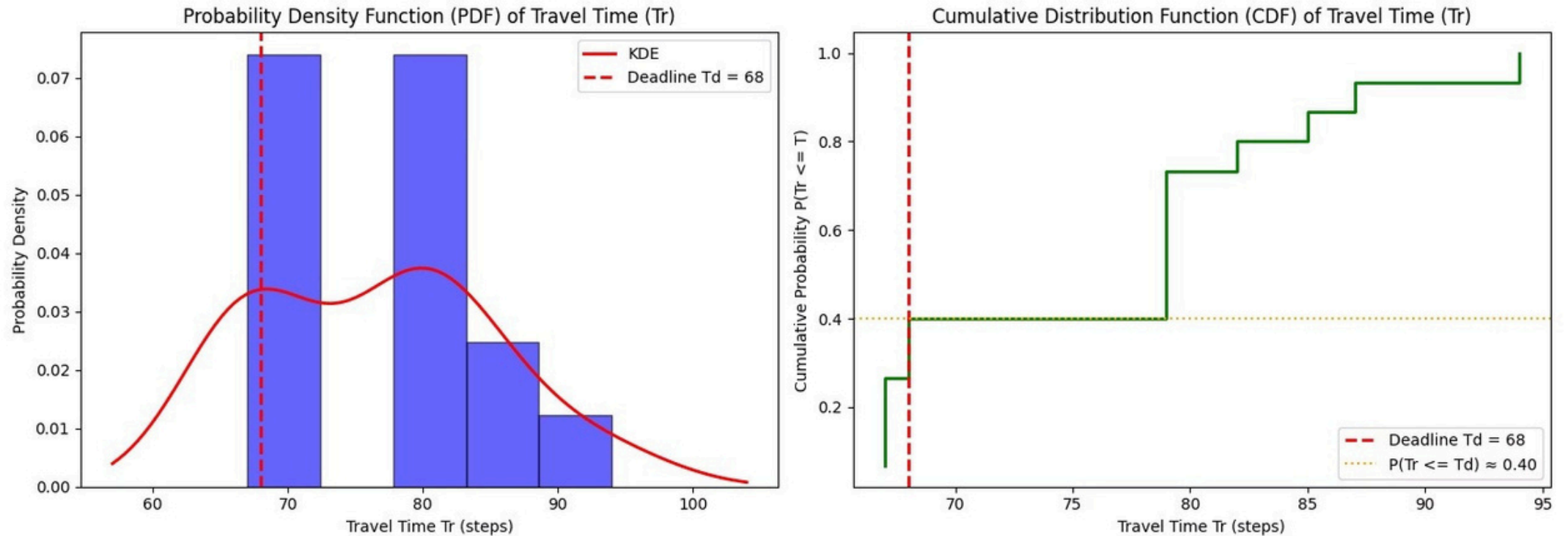
Sum segment times: $T_{route}(k) = \sum T(i)_k$



Compare Deadline

Evaluate $T_{route}(k)$ against deadline D

Simulation Results and Analysis



Generated Distributions

- Probability DensityFunction (PDF) of travel times
- Cumulative Distribution Function (CDF)
- Arrival time deviation relative to deadline

Reliability Metric

CDF estimates $P(\text{Troute} < D)$ probability of on-time arrival



Limitations

Limitations of Deterministic Strategy

- **Ignores Travel Time Variability:** Route selection is based only on expected travel time and does not consider variance or reliability of travel times.
- **Fixed Routing Strategy :** The route computed using the A* algorithm remains unchanged across all simulation runs, even when traffic conditions vary.
- **No Risk Awareness :** The deterministic approach minimizes expected travel time but does not minimize the probability of missing the deadline.
- **Simplified Traffic Modeling :** The model assumes independence between road segments, while in reality traffic conditions are often correlated.
- **No Dynamic Adaptation :** The routing decision does not update in response to new traffic information or unexpected congestion.
- **Single Path Decision :** The system always selects one fixed path instead of considering multiple alternative routes under uncertainty.

Summary and Transition to Randomized Strategy

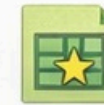


Deterministic

- Ignores Variability
- Fixed Route
- No Risk Awareness



Always Same Route



Randomized

- Adapt to Uncertainty
- Probabilistic Choices
- Risk-Aware Decision



Flexible Routes

The **deterministic approach** is limited. Switch to a randomized strategy that adapts to uncertainty and **minimizes delay risk**.